Spin-Polarized Fusion Neutronics

A from-scratch primer on everything we discussed

(assumes: multivariable calculus, linear algebra, intro upper-division physics)

built for the train ride home

July 8, 2026

How to read this

- Each part starts from something you already know and builds one step at a time.
- **Blue** words are the terms worth remembering.
- Formulas are here, but every one comes with a plain-English sentence. If you skip the algebra, read the sentence.
- Goal: by the end you can explain the whole project to a friend — the physics, the engineering, the computation, and why any of it matters.

The one-sentence version

If you line up the spins of fusion fuel, the neutrons come out in a preferred direction — and we built the first tool inside a standard reactor-physics code to simulate that, then figured out when it actually helps.

Three worlds meet here:

- **Physics**: why aligned spins make neutrons directional.
- **Engineering**: why a reactor designer cares where neutrons land.
- **Computation**: how you simulate it honestly, and where cheap shortcuts lie to you.

Part 1

Physics refreshers: fusion, spin, and magnetic confinement.

Fusion, and the neutron we care about

The reaction in a D–T reactor:

$$\underbrace{{}^{2}\mathrm{D}}_{\text{deuteron}} + \underbrace{{}^{3}\mathrm{T}}_{\text{triton}} \rightarrow \underbrace{{}^{4}\mathrm{He}}_{3.5\,\mathrm{MeV}} + \underbrace{n}_{\mathbf{14.1\,MeV}}$$

- The **neutron** carries 14.1 MeV ($1\,\mathrm{MeV} = 1.6 \times 10^{-13}$ J). It is **uncharged**, so magnetic fields do *not* bend it — once born it flies straight until it hits something.
- That neutron is the entire product we track: it breeds tritium, heats the blanket, and damages the walls. Everything downstream is "where do the neutrons go?"

Spin, in one slide (quantum refresher)

Spin is an intrinsic angular momentum of a particle — think of it as a tiny built-in arrow that, in a magnetic field, can only point in a few discrete directions.

- The **deuteron** has spin 1: its projection on $\hat{\mathbf{B}}$ can be $+1$, 0, -1 (call the fractions d_+, d_0, d_-).
- The **triton** has spin $\frac{1}{2}$: projection $+\frac{1}{2}$ or $-\frac{1}{2}$ (fractions t_+, t_-).

Normally the fuel is **unpolarized**: all projections equally likely, arrows point every which way. **Polarizing** the fuel means preparing the spins so specific projections dominate. That is a knob a reactor designer could, in principle, turn.

Magnetic confinement (E&M refresher)

A hot fusion plasma is charged (ions + electrons). From the Lorentz force $\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}$, a charged particle **spirals around magnetic field lines** — free to move along $\hat{\mathbf{B}}$, tightly bound across it.

- So if you can make field lines that close on themselves inside a donut (torus), you trap the plasma. That is a magnetic confinement device.
- The field lines wrap around on nested **flux surfaces** — like the layers of an onion. Each surface has (roughly) constant temperature and density.
- Key point for us: **at every point there is a well-defined field direction $\hat{\mathbf{B}}$**. The polarized neutrons are emitted relative to *that local* $\hat{\mathbf{B}}$.

Two ways to build the donut

Tokamak

- Axisymmetric: looks the same at every toroidal angle ϕ.
- Needs a big electric current in the plasma to twist the field.
- Simple shape; the current can cause disruptions.

Stellarator

- Fully 3D: the cross-section twists and changes as you go around.
- The twist is built into fancy 3D coils — no plasma current needed, so it is steadier.
- Much harder to design and build; the geometry is genuinely three-dimensional.

Axisymmetry ("same at every ϕ") is the single most important distinction for us: it is what makes a tokamak easy to compute and a stellarator hard.

Part 2

Why aligned spins make neutrons come out sideways.

The cross section: how likely, and in what direction

A **cross section** σ measures how likely a reaction is (units of area — a bigger "target"). The **differential cross section** $\dfrac{d\sigma}{d\Omega}$ adds *direction*: how likely to emit into a small solid angle $d\Omega$ around some direction.

- **Solid angle** Ω = the 3D version of an angle; the full sphere is 4π.
- For unpolarized fuel, emission is **isotropic**: same in every direction, $\frac{d\sigma}{d\Omega} = \frac{\sigma}{4\pi}$.
- For *polarized* fuel, it is not — and that is the whole point.

The master formula (Schwartz 2025)

With θ_B = angle between the emitted neutron and the local field $\hat{\mathbf{B}}$:

$$\frac{d\sigma}{d\Omega} = \frac{\sigma_0}{2\pi}\left[\tfrac{3}{4}\,a\,\sin^2\theta_B + \left(\tfrac{2}{3}b + \tfrac{1}{3}c\right)\left(\tfrac{1}{4} + \tfrac{3}{4}\cos^2\theta_B\right)\right]$$

The three numbers **(a,b,c)** are set by how you polarized the fuel, and $a+b+c=1$:

$$a = d_+t_+ + d_-t_-, \qquad b = d_0, \qquad c = d_+t_- + d_-t_+.$$

Unpolarized fuel is $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$. Read the formula as: **a bit of "$\sin^2$" shape plus a bit of "$\cos^2$" shape**, weighted by how you prepared the spins.

What $\sin^2\theta_B$ and $\cos^2\theta_B$ look like

- $\sin^2\theta_B$: **zero along $\hat{\mathbf{B}}$, maximum perpendicular** to it. Neutrons squirt out sideways, in the plane perpendicular to the field. Call this the **A / "perp"** shape.
- $\frac{1}{4}+\frac{3}{4}\cos^2\theta_B$: **maximum along $\hat{\mathbf{B}}$**, minimum sideways. Neutrons prefer to go along the field. Call this the **B/C / "par"** shape.

Legendre reminder: $\cos^2\theta = \frac{1}{3}(1+2P_2(\cos\theta))$ where $P_2(x)=\frac{1}{2}(3x^2-1)$ is the 2nd Legendre polynomial. So both shapes are "isotropic plus a P_2 wobble."

Two knobs, not one — keep them separate

The formula secretly contains **two independent things**:

1. **Direction (shape).** Which way the neutrons prefer to go. Controlled by one number
$$w(\theta_B) \propto 1 + a_2\, P_2(\cos\theta_B), \qquad a_2 \in [-1, +1].$$
$a_2 = -1$: pure perp. $a_2 = +1$: pure par. $a_2 = 0$: isotropic.
2. **Rate (how many).** The total number of neutrons, via $\eta = a + \frac{2}{3}b + \frac{1}{3}c$.

Design lesson: in code, sample the *direction* from the pdf and multiply the *count* by η separately. Mixing them is the classic bug.

The three modes, in plain words

Mode	Rate η	What it does
Unpolarized	$2/3$	baseline, neutrons everywhere
A (perp)	$1\ (+50\%)$	most power; neutrons sideways to $\hat{\mathbf{B}}$
B (par)	$2/3$ (same)	same power; neutrons along $\hat{\mathbf{B}}$
C (par)	$1/3\ (-50\%)$	half the power; same direction as B

Surprise (and a correction to common lore): B and C point the *same* way — they differ only in *how many*. C is *not* isotropic. The $+50\%$ reactivity boost (Kulsrud 1982) lives in mode A.

Part 3

Flux surfaces, VMEC, and why stellarators are hard.

Describing the plasma shape: flux coordinates

Instead of (x, y, z) we use **flux coordinates** (ρ, θ, ζ):

- ρ: which onion layer (0 = center, 1 = edge / **last closed flux surface**).
- θ: the short way around the donut (poloidal angle).
- ζ: the long way around the donut (toroidal angle).

A computer code called **VMEC** solves the plasma equilibrium and hands you, for every (ρ, θ, ζ): the real position (R, Z, ϕ), the field direction $\hat{\mathbf{B}}$, and the **Jacobian** $\sqrt{g}$.

What $\sqrt{g}$ is, and why it matters (multivar calc)

When you change variables in an integral you multiply by the Jacobian determinant:

$$\int f\,dx\,dy\,dz = \int f \underbrace{\sqrt{g}}_{\text{Jacobian}} d\rho\,d\theta\,d\zeta.$$

- $\sqrt{g}$ tells you how much *real volume* sits in each little (ρ, θ, ζ) cell. On a distorted 3D flux surface some cells are fat, some thin.
- To place fusion births correctly you must weight by **reactivity** $\times\sqrt{g}$ — otherwise you put too many neutrons where the coordinate grid is dense but the real volume is small. Getting this right is a big part of "rigorous."

Quasi-symmetry: QA and QH

Stellarators are 3D, but a clever subclass **hides a hidden symmetry in the field strength** $|\mathbf{B}|$ (not in the shape). Particles then confine almost as well as in a tokamak.

- **QA** (quasi-axisymmetric): $|\mathbf{B}|$ looks tokamak-like.
- **QH** (quasi-helical): $|\mathbf{B}|$ has a helical symmetry.
- **nfp** = number of field periods (the shape repeats nfp times around).
- ι (iota) = rotational transform = how much a field line twists poloidally per toroidal lap.

Database used: **QUASR**, a public catalogue of $\sim$370,000 optimized stellarators.

The killer subtlety: symmetry $\neq$ gentle shape

Good quasi-symmetry is bought with strong 3D boundary shaping.

So the devices with the best physics ($|\mathbf{B}|$ symmetry) are often the most geometrically distorted. This anti-correlation is why “the plasma looks almost axisymmetric” is a bad predictor of anything — and it foreshadows why cheap analytic methods struggle exactly on the good devices.

We measure geometric distortion with ε_{eff}:

$$\varepsilon_{\text{eff}} = \frac{\text{helical excursion of the source}}{\text{source-to-wall standoff}}.$$

Small = gentle, large = strongly shaped. Real optimized QAs sit at $\varepsilon_{\text{eff}} \approx 1$–$2.6$.

Part 4

Neutron wall loading, breeding, and why anyone cares.

Where the neutrons land: the first wall

- The **first wall** is the innermost solid surface facing the plasma. Behind it sits the **blanket**.
- **Neutron wall loading (NWL)** = neutron power hitting the wall per unit area ($\mathrm{MW/m^2}$). Too much $\Rightarrow$ the wall erodes, the magnets behind it get damaged.
- **Peaking factor** $\mathrm{PF} = \dfrac{\max \mathrm{NWL}}{\text{average NWL}}$. $\mathrm{PF} = 1$ is perfectly uniform; big PF means a nasty hot spot. Designers want PF low.
- **Inboard** = the inner (small-radius, "center-stack") wall, usually the tightest, most shielding-starved real estate. **Outboard** = outer wall, roomier.

Breeding: the reactor must make its own fuel

Tritium is rare, so the reactor breeds it in the blanket:

$$n + {}^{6}\mathrm{Li} \rightarrow {}^{4}\mathrm{He} + \mathrm{T} + 4.8\,\mathrm{MeV}.$$

- **TBR** (tritium breeding ratio) = tritium atoms bred per fusion neutron. You need **TBR** > 1 (about 1.15 with margin) or the reactor starves.
- **FLiBe** = a molten salt (Li, Be, F) used as breeder+coolant; **Be** multiplies neutrons via $(n, 2n)$.
- Our finding: TBR *per neutron* barely changes with polarization, so steering the neutrons around does *not* break breeding self-sufficiency. Good news.

The dream: steer neutrons with spin

If polarization aims neutrons, maybe we can aim them away from the fragile inboard wall and lightly-shielded magnets — buying component lifetime for free.

This is the motivating hope, and it is exactly what SPF does in spherical tokamaks (Bae 2025: +68% magnet lifetime). The scientific question of this project: **does it also work for stellarators, and how much of the promised benefit is real once you account for everything?**

Part 5

Simulating neutrons honestly.

Monte Carlo transport in 90 seconds

Monte Carlo = simulate millions of individual neutrons with random numbers, then average. **OpenMC** is the open-source code we use.

- Each neutron is **born** (the "source"), then flies, maybe **scatters**, maybe gets absorbed, until it dies. Tally what hits the wall.
- **Free-streaming / first-flight**: pretend neutrons never scatter — they fly straight from birth to wall. A useful idealization that isolates the source geometry.
- **Statistical error** shrinks like $1/\sqrt{N}$: to resolve a 1% effect you need $\sim 10^4$ counts *per bin*. This is why small effects are expensive in MC.

The “source” is the thing we built

The **source** decides where each neutron is born and which way it points. Ours must:

1. pick a birth position (weighted by reactivity $\times\sqrt{g}$);
2. look up the local field $\hat{\mathbf{B}}$ there;
3. sample a direction from $w(\theta_B) \propto 1 + a_2 P_2(\cos\theta_B)$ *about that* $\hat{\mathbf{B}}$;
4. rotate that direction into lab coordinates.

Doing step 2–3 **on the fly, per neutron, in the local field frame** is what makes it “native” and general — and is why it works for a twisting stellarator field where $\hat{\mathbf{B}}$ changes everywhere.

How you sample a direction (the computation)

- **Inverse-CDF sampling**: to draw from a pdf $p(x)$, invert its cumulative $u = \int_{-\infty}^{x} p$; feed a uniform random $u \in [0, 1)$, get x. Exact and fast.
- **Rejection sampling**: throw darts under an easy envelope, keep those under p. Slower but foolproof — we use it to *cross-check* the inverse-CDF.
- **Stratified sampling**: split the mixture (perp vs par) and sample each exactly — gives exact draws and makes per-mode testing trivial.
- **Thread safety**: OpenMC runs many neutrons in parallel threads. If the source stored mutable state it would corrupt. Rule: set everything at construction, then **never write** — "`const`."

A verification trick worth stealing: linearity in a_2

Because P_2 enters *linearly*, the wall load is exactly linear in a_2:

$$\mathrm{NWL}(a_2) = \mathrm{NWL}_{\mathrm{unpol}} + a_2\big(\mathrm{NWL}_{\mathrm{par}} - \mathrm{NWL}_{\mathrm{unpol}}\big).$$

Set $a_2 = +1$ and $a_2 = -1$ and add:

$$\boxed{\mathrm{NWL}_{\mathrm{perp}} + \mathrm{NWL}_{\mathrm{par}} = 2\,\mathrm{NWL}_{\mathrm{unpol}}}$$

This must hold **bin by bin**, with *no analytic reference needed.* If it fails, the sampler is wrong. Cheap, exact, self-checking — the kind of test reviewers trust.

Part 6

The results — and the honest caveats.

Finding 1: where the cheap analytic method lies

The competing cheap method treats each source filament and computes flux to a wall patch by a **point-patch** formula. On a real QA it agrees with MC to $\leq 1.8\%$ *except* on the sharply curved inboard wall (4–6% off).

- A controlled test shows the analytic error grows like $(\textbf{shaping})^2$ and is $\sim 3\times$ worse on the convex inboard wall.
- At realistic shaping ($\varepsilon_{\text{eff}} \approx 2.3$) the analytic *quadrature itself* breaks (34% error, negative loads). **This regime is Monte-Carlo-only.**

That is the intellectual justification for using a transport code at all.

Finding 2: scattering dilutes the steering $\sim 3\times$

The single most important honesty result.

- Free-streaming says polarization shifts the inboard load by $\pm 40\%$.
- Put in a *real blanket* with scattering on, and it is only ± 13–15%.
- Why: neutrons that enter the wall **scatter back** into the cavity as a direction-blind diffuse fog, washing out the aim.

Geometric and analytic steering numbers are **optimistic by $\sim 3\times$**, and *only a full transport calc reveals it.* The analytic theory structurally cannot.

Finding 3: rate vs steering — "B is the sweet spot"

Restoring the true neutron count η per mode (vs unpolarized $= 1$):

	fusion rate	center-stack load	outboard	TBR/neutron
A	1.50	**1.70**	1.36	1.14
B	1.00	**0.85**	1.09	1.16
C	0.50	0.42	0.55	1.16

- A: more power, but aims *into* the center stack $\Rightarrow$ load compounds to $1.70\times$.
- **B: relieves the center stack ($0.85\times$) at no power or breeding cost** — the free lunch.
- C: relieves the most, but you paid half your power for it.

Finding 4: strong local actuator, weak global flattener

Polarization moves a lot of load locally (40–70% control at a point) but barely lowers the overall peaking factor.

Why: the peaking is set by a narrow **near-field hot spot** (the wall's closest approach to the plasma). Angular redistribution reshapes the broad load but **cannot move that spike**.

- Example device: optimal polarization drops PF only $1.78 \to 1.49$.
- **Two independent methods agree** (our MC *and* the analytic companion): SPF is a scalpel, not a flattening iron.

This reframes what SPF is *for*: local fluence tailoring on critical parts, not global peaking reduction.

Part 7

anarrima: doing it with calculus instead of dice.

Why an analytic method at all

Monte Carlo is noisy and slow for small effects and gives no exact derivatives. An analytic formula is instant, noise-free, and **differentiable** (you can optimize with it). The catch: it only works where the geometry is gentle. anarrima is that analytic method for polarized NWL.

Near-singular integrals and "singularity subtraction"

The wall-load integral looks like $\int \frac{f(\phi)}{D(\phi)^{3/2}}\, d\phi$, where D = squared distance from source to wall. When the wall nearly grazes the source, $D \to 0$ and the integrand spikes — ordinary numerical integration (**Gauss–Legendre quadrature**) needs thousands of points and still fails.

- **Trick**: because $D(\phi)$ is a trig polynomial, you can find the spike's location, depth, and curvature **in closed form**, subtract an exactly-integrable model of it, and integrate the smooth leftover with **12 points** instead of 6000.
- Kept **differentiable**, so you still get exact gradients. This is the freshest numerical-methods contribution.

Autodiff, elliptic integrals, and the two thresholds

- **Automatic differentiation (autodiff / JAX)**: the computer differentiates your code exactly, for free — gold for optimization. anarrima is built to be autodiff-clean.
- **Carlson elliptic integrals**: the special functions the toroidal geometry produces; we implement them with exact hand-coded derivatives.
- **Two $\varepsilon_{\mathrm{eff}}$ thresholds**: the *series* converges only for $\varepsilon_{\mathrm{eff}} \lesssim \frac{1}{2}$; the *quadrature* survives to ~ 2. They are different numbers — a genuine, citable clarification.
- **$\mathbf{gc}_2$ correction**: to compute emission about the *unit* field $\hat{\mathbf{B}} = \mathbf{B}/|\mathbf{B}|$, you must differentiate the normalization $|\mathbf{B}|$ too; the original derivation dropped a 2nd-order piece. Restoring it pushes the polarized series from 2nd to 3rd order accuracy. (That was your "gc2 improvement.")

A clean side result: the wall is a low-pass filter

Question: does tokamak **field ripple** (the field wobble from having a finite number N of coils) show up on the wall?

$$\text{transmission of harmonic } N \;\sim\; e^{-N\,a/R_0}.$$

- The line-of-sight integration **blurs out** high-frequency source structure. For a real tokamak $Na/R_0 \approx 6$, so the ripple is suppressed $\sim$100× — the wall signal is $\sim$50 parts per million. Negligible.
- **Bonus insight**: this same blurring is *why* "plasma shape" metrics failed to predict wall behavior — the wall literally cannot see high-harmonic shape. One mechanism explains several dead ends.

Part 8

What we tried to predict, and what honesty demands.

The hunt for a cheap predictor

Dream: a single number from the plasma shape that predicts SPF benefit, so you skip the expensive transport. Attempts:

- **Nematic order parameter** Q (borrowed from liquid-crystal physics: how aligned a set of direction vectors are) — **did not predict**.
- A "C-term" shape metric — wrong order, **did not predict**.
- **Field-direction coherence** (mean resultant length of $\hat{\mathbf{B}}$ in the right frame): sharper — it cleanly separates QA from QH, but the scalar alone is weak.

Best current predictor is unglamorous: **available headroom** — devices with a big hot spot have more to gain. And the low-pass slide explains *why* shape metrics can't work.

Statistics honesty (why n matters)

- **Spearman correlation** ρ: rank-based measure of "do these two move together."
- With only $n = 5$ devices, $\rho = \pm 0.6$ is *within noise* — you cannot conclude anything. We said so, and launched a bigger batch ($n \approx 25$) to actually decide.
- **Statistical power**: the ability to detect a real effect. Small n = low power = easy to fool yourself. Reporting the null honestly ("we could not tell") is a result, not a failure.

This honesty — reporting the $\sim 3\times$ dilution, the failed predictors, the n-too-small caveats — is what makes the work trustworthy.

Part 9

How it all becomes one story.

What is genuinely new (and what is not)

- **New**: first *native, on-the-fly, general* polarized fusion source in a production MC code; first *polarized stellarator* source; first stellarator source *in OpenMC*.
- **Not first**: “a polarized source in OpenMC” (Bae 2025, but precomputed, one tokamak); “a VMEC stellarator source” (Lyytinen 2024, but Serpent2, isotropic, geometry- focused). We cite these up front and differentiate.
- **Precedent**: Peterson’s tokamak-source paper shows a native+validated tool is publishable even as an increment — and we clear that bar more easily (no prior stellarator source in OpenMC; we add polarization).

The paper's spine

A native spin-polarized source for OpenMC — and where/why Monte Carlo is necessary for polarized stellarator wall loading.

1. Physics + the native sampler.
2. StellaratorSource from a verified equilibrium.
3. Verification (incl. the linearity identity).
4. Where MC is necessary: analytic breakdown; the $3\times$ scattering dilution; rate-vs-steering; the weak-global/strong-local finding.
5. Analytic companion (anarrima): singularity subtraction, differentiable quadrature, ε_{eff} thresholds — and it independently confirms the main finding.

Genre: a **methods + honest-findings** paper. It does not need a blockbuster discovery; it needs a correct, validated, useful capability, characterized honestly.

The five things to remember

1. Aligned spins $\Rightarrow$ directional neutrons: one knob for **direction** (a_2), one for **rate** (η).
2. You need the **real local field $\hat{\mathbf{B}}$** from a **real equilibrium** — that is what makes the stellarator source rigorous and the polarization well-posed.
3. **Scattering dilutes steering** $\sim 3\times$; only transport shows it.
4. SPF is a **local scalpel, not a global flattener** — two methods agree.
5. Cheap analytic methods **break where the geometry is strong**; that boundary is the paper's thesis.

Enjoy the train ride.

Questions this deck should now let you answer:
Why do polarized neutrons care about $\hat{\mathbf{B}}$*? Why is a stellarator harder than a tokamak? Why is free-streaming optimistic? What is a peaking factor, and why can't polarization fix it? What does "singularity subtraction" buy you?*